

Assignment – 1

Term - 1, A+ - Understanding Of Hardware And Its Components

Section 1: Multiple Choice

1. Which of the following is NOT a component of the CPU?

-> 2/RAM

2. What is the function of RAM in a computer?

-> RAM is Temporary and volatile memory. In RAM Stored Data Get Lost When We Shut Down Computer.

-> There Are Two Types Of RAM.

i)SRAM(Static RAM)

ii)DRAM(Dynamic RAM)

1)S.I.M.M.(Single Inline Memory Module)

2)R.I.M.M.(Rambus Inline Memory Module)

3)D.I.M.M.(Duel Inline Memory Module)Now In Use..

3. Which of the following is a primary storage device?

1. HDD

2. SSD

3. SD card

4. 1 and 2 both

-> None Of This Are Primary Storage Device.

4. What is the purpose of a GPU?

-> GPU (Graphic Processing Unit) work is to Graphic Rendering and Parallel Processing. It Enhances gaming performance.

5. True or False: The motherboard is the main circuit board of a computer where other components are attached.

-> True.

6. True or False: A UPS (Uninterruptible Power Supply) is a hardware device that provides emergency power to a load when the input power source fails.

-> True.

7. True or False: An expansion card is a circuit board that enhances the functionality of a component.

-> True.

8. Explain the difference between HDD and SSD.

-> The Main Deference between HDD And SSD is That SSD Is More Faster And Durable Than HDD.

9. Describe the function of BIOS in a computer system.

-> Bios Have 4 Functions:-

1.POST (Power On Self test)

2.Boot Loader

3.Bios Driver

4. CMOS

10. List and briefly explain three input devices commonly used with computers.

-> Mouse:- Mouse Is One Of The Input Devices. It works On Cursor by Pointing and clicking.

Keyboard:- We Can Type On key Board And Give Input To computer by this.

Microphone or mike :- By Speaking In It We can provide an input to computer.

11. Identify and label the following components on a diagram of a motherboard:

- CPU**
- RAM slots**
- SATA connectors**
- PCI-E slot**

12. Demonstrate how to install a RAM module into a computer.

13. Discuss the importance of proper cooling mechanisms in a computer system. Include examples of cooling methods and their effectiveness.

->Cooling Mechanisms Are Essential for Computer's Health ,Performance And Stability. Because Overheating Can Slow Down, Crashes To Computer.

->The Reason Why Cooling Is Important.

- Because Cooling System Prevents Overheating.**
- Reduce Physical damage from heat.**
- prevent crashes and shutdown from overheating.**
- By Reducing Heat it boosts Performance.**

->Cooling Methods And Their Effectiveness.

- Air Cooling method use fan to cooldown GPU and CPU.**

And it is cost Effective but for high performance.

-Liquid Cooling method Uses coolant in tubes to transfer heat away from the CPU/GPU. And It more effective than air cooling and more for gaming & High pc.

-Phase Change Cooling Uses Too Cool components for cooling too zero temperature. And It is Too Effective But A lot expansive and complex.

14. Explain the concept of bus width and its significance in computer architecture.

-> A bus in computer architecture is a communication pathway that transfers data between different components of a computer, such as the CPU, memory (RAM), and storage devices.

-> There Is Three Type Of System Buses:-

1)Control Bus

2)Address Bus

3)Data Bus

1)Control Bus

- To Transfer Control Signal From One Component To Other

2)Address Bus

-To Carry Address Signals to Read And Write Data In memory

3)Data Bus

-It is Used To Data Transfer One From Other.